
The Progression Masterclass

Topic 1: Arithmetic Progressions

This is the foundation. An **Arithmetic Progression (AP)** is a sequence where each term is found by adding a constant “common difference” (d) to the one before it. It is one of the simplest, yet most fundamental, types of sequences.

AP Formulas

- n -th term (a_n): $a_n = a + (n - 1)d$
- Sum of the first n terms (S_n), where l is the last term: $S_n = \frac{n}{2}(a + l)$
- Sum using the first term and common difference: $S_n = \frac{n}{2}[2a + (n - 1)d]$

Practice Problem

The sum of the first three terms of an AP is 27, and the sum of their squares is 293. Find the three terms.

(Hint: Let the three terms be $a - d, a, a + d$. Their sum is $3a = 27$, which gives you the middle term immediately. Then use the sum of squares to find the common difference d .)

Advanced Challenge

A man saves \$100 in his first month and then increases his savings by \$25 each subsequent month. In how many months will his total savings exceed \$5000 for the first time?

Topic 2: Geometric Progressions

A **Geometric Progression (GP)** is a sequence where each term is found by multiplying the previous one by a constant “common ratio” (r).

GP Formulas

- n -th term: $a_n = a \cdot r^{n-1}$
- Sum of a finite GP: $S_n = \frac{a(1-r^n)}{1-r}$ (for $r \neq 1$)
- Sum of an infinite GP: $S_\infty = \frac{a}{1-r}$ (only if $-1 < r < 1$)

Practice Problem

The 3rd, 6th, and 12th terms of an Arithmetic Progression are consecutive terms of a Geometric Progression. If the first term of the AP is not zero, find the common ratio (r) of the GP.

(Hint: Let the AP have first term a and common difference d . The terms are $a + 2d, a + 5d, a + 11d$. If they are in a GP, then the defining property is $(a + 5d)^2 = (a + 2d)(a + 11d)$. Solve this equation for d in terms of a , then find the common ratio.)

Advanced Challenge

A ball is dropped from a height of 81 feet. Each time it strikes the ground, it rebounds to two-thirds of the height from which it fell. Find:

1. The height of the ball at the top of its 5th bounce.
2. The total distance the ball travels before coming to rest.

Topic 3: The Method of Finite Differences

What if a sequence isn't an AP or GP? Some sequences follow a polynomial rule (e.g., $a_n = n^2 + 3n - 1$). The **Method of Finite Differences** is a powerful tool to find this rule. You write the sequence, then write the sequence of differences below it, then the differences of the differences, and so on, until you get a constant row. If it takes k rows of differences to get a constant, the general term is a polynomial of degree k .

Practice Problem

Find the general formula (the n -th term) for the sequence: 4, 15, 32, 55, 84, ...

(Hint: Write the sequence in a row. Below it, write the differences between consecutive terms ($15 - 4, 32 - 15$, etc.). Is this new sequence an AP? If not, repeat the process. The number of steps it takes to reach a constant difference tells you the degree of the polynomial.)

Advanced Challenge

A “tetrahedral number”¹ is the number of spheres in a triangular pyramid (a tetrahedron). The sequence is 1, 4, 10, 20, 35, ... Use the method of finite differences to find a formula for the n -th tetrahedral number.

¹ T_n is the n -th triangular number, $T_n = \frac{n(n+1)}{2}$. The tetrahedral numbers are given by $\sum T_n = \frac{n(n+1)(n+2)}{6}$.

Topic 4: Recurrence Relations

Some sequences are defined by their relationship to previous terms. The most famous is the **Fibonacci sequence** ($F_n = F_{n-1} + F_{n-2}$). These are called recurrence relations. The key is to understand how the sequence builds upon itself. Many combinatorics (counting) problems secretly turn into recurrence relations.

A sequence is defined by a formula that connects a_n to its predecessors, like a_{n-1}, a_{n-2} , etc.

Practice Problem

A sequence is defined by $a_1 = 2$ and $a_{n+1} = 2a_n + 1$ for $n \geq 1$. Find the value of a_{10} .

(Hint: Calculate the first few terms: a_1, a_2, a_3, a_4 . A direct pattern might be hard to spot. Try creating a new sequence $b_n = a_n + 1$. Write out the first few terms of b_n . Is this new sequence familiar? Once you find a formula for b_n , you can easily find the formula for a_n .)

Advanced Challenge

Let a_n be the number of binary strings of length n that do not contain two consecutive 0s. Find the value of a_8 .

Topic 5: Summation of Series (Σ Notation)

Often, we need to find the sum of a sequence. **Sigma (Σ) notation** is the compact language for this. We will learn how to use it and, most importantly, learn the formulas for the sum of the first n integers, squares, and cubes.

$$\begin{aligned} \bullet \sum_{k=1}^n k &= \frac{n(n+1)}{2} \\ \bullet \sum_{k=1}^n k^2 &= \frac{n(n+1)(2n+1)}{6} \\ \bullet \sum_{k=1}^n k^3 &= \left(\frac{n(n+1)}{2} \right)^2 \end{aligned}$$

Practice Problem

Find the sum of the first 10 terms of the series whose n -th term is given by $a_n = n(n+2)$. That is, calculate $\sum_{k=1}^{10} (k^2 + 2k)$.

(Hint: Use the properties of sigma notation: $\sum(A+B) = \sum A + \sum B$ and $\sum(cA) = c \sum A$. Split the sum into two separate, simpler sums that you have formulas for.)

Advanced Challenge

Find a closed-form formula in terms of n for the sum of the series: $1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \cdots + n(n+1)(n+2)$.

Topic 6: Telescoping Series

This is a beautiful and clever trick. A **telescoping series** is one where, upon expanding the terms, most of them cancel each other out. The key is to rewrite the terms of the series as a difference of two expressions, typically using partial fraction decomposition.

Practice Problem

Calculate the exact value of the sum: $S = \frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \frac{1}{7 \cdot 10} + \cdots + \frac{1}{28 \cdot 31}$.

(Hint: Try to write a general term $\frac{1}{(3k-2)(3k+1)}$ as a difference. Using partial fractions, you'll find it is equivalent to $\frac{1}{3} \left(\frac{1}{3k-2} - \frac{1}{3k+1} \right)$. Now write out the first few terms of the sum using this new form and see what cancels.)

Advanced Challenge

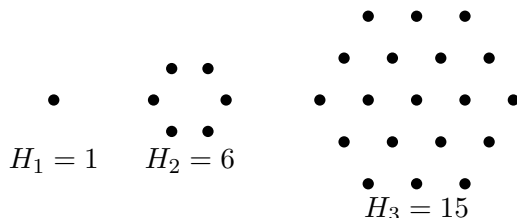
Find the sum $S = \frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{99} + \sqrt{100}}$.

Topic 7: Figurate Numbers (Patterns You Can See)

Figurate numbers are sequences represented by geometric arrangements of dots. The most famous are **Triangular Numbers** ($T_n = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$), **Square Numbers** ($S_n = n^2$), and Pentagonal Numbers. This connects algebra with geometry.

Practice Problem

Find a formula for the n -th hexagonal number. The first few are 1, 6, 15, 28, ... Here is a visual hint showing the first three hexagonal numbers:



(Hint: How is the n -th hexagon built from the $(n-1)$ -th one? Alternatively, use the Method of Finite Differences on the sequence 1, 6, 15, 28. A third way: can you see a relationship between the n -th hexagonal number and the n -th triangular number?)

Advanced Challenge

The “Square Pyramidal Numbers” are the sum of the first n squares: $P_n = \sum_{k=1}^n k^2$. There is a beautiful geometric proof that shows $P_n = \frac{n(n+1)(2n+1)}{6}$. Find a relationship between P_n , the triangular number T_n , and T_{n-1} .

Topic 8: Arithmetico-Geometric Progressions (AGP)

An AGP is a blend of an AP and a GP. Each term is the product of a term from an AP and a term from a GP. For example: $1 \cdot x, 3 \cdot x^2, 5 \cdot x^3, \dots$. Finding the sum of these series requires a very clever and famous trick.

Practice Problem

Find the sum of the infinite series: $S = 1 + \frac{2}{3} + \frac{3}{9} + \frac{4}{27} + \frac{5}{81} + \dots$

(Hint: The trick: multiply the entire sum S by the common ratio of the geometric part (which is $1/3$). Write out S and $\frac{1}{3}S$ one below the other, but shift the terms of $\frac{1}{3}S$ one place to the right. Now, subtract the two series ($S - \frac{1}{3}S$). You will find that most of the terms align to form a simple GP.)

Advanced Challenge

A company buys a new server for \$10,000. Each year, they add more servers costing \$1,000 more than the previous year's addition (first addition: \$1,000, second: \$2,000, etc.). However, each server's value depreciates by 20% annually. What is the total value of their server inventory after 5 years?

Topic 9: Sequences in Counting Problems

Many “How many ways...?” problems in combinatorics can be solved by defining a sequence. Let a_n be the number of ways to do something for a size n problem. Then, try to find a relationship between a_n and smaller terms like a_{n-1} or a_{n-2} . This creates a recurrence relation.

Practice Problem

A staircase has n steps. You can climb either 1, 2, or 3 steps at a time. In how many distinct ways can you climb to the top of a staircase with 10 steps?

Formulate a recurrence relation for the number of ways, and compute the answer for $n = 10$.

Advanced Challenge

Let a_n be the number of ways to color the squares of a $1 \times n$ board using red, blue, and green, such that no two adjacent squares are the same color and no three consecutive squares are all different colors.

- (a) Find a recurrence relation for a_n .
- (b) Compute a_5 .

Topic 10: Harmonic Progressions

A sequence is a **Harmonic Progression (HP)** if the reciprocals of its terms form an Arithmetic Progression (AP). For example, $1, 1/2, 1/3, 1/4, \dots$ is an HP because $1, 2, 3, 4, \dots$ is an AP. This leads to the powerful **AM-HM Inequality**. For any two positive numbers a and c , their Arithmetic Mean is always greater than or equal to their Harmonic Mean:

$$\frac{a+c}{2} \geq \frac{2}{\frac{1}{a} + \frac{1}{c}}$$

Practice Problem

The 4th term of a Harmonic Progression is 6, and the 7th term is 12. What is the 9th term?

(Hint: Don't work with the HP directly. Convert it into an AP problem! If h_n are the terms of the HP, let $a_n = 1/h_n$. You are given $h_4 = 6$ and $h_7 = 12$. What does this tell you about a_4 and a_7 ? Use this information to find the common difference of the AP, then find a_9 , and finally convert back to h_9 .)

Advanced Challenge

If x, y, z are in harmonic progression, show that $\frac{x}{y} + \frac{y}{x} + \frac{x}{z} + \frac{z}{x} = \frac{y}{z} + \frac{z}{y} + 4$.

Topic 11: The Gauss Trick and Symmetry

Sometimes, the most elegant solutions in mathematics come from looking at a problem from a different angle. The story of young Carl Friedrich Gauss, tasked with summing the integers from 1 to 100, is a perfect example of this. Instead of brute force, he used symmetry.

He wrote the sum down, and then wrote it again in reverse order below the first:

$$\begin{aligned} S &= 1 + 2 + 3 + \cdots + 99 + 100 \\ S &= 100 + 99 + 98 + \cdots + 2 + 1 \end{aligned}$$

Adding these two lines vertically, each pair of terms sums to 101:

$$2S = (1 + 100) + (2 + 99) + \cdots + (100 + 1) = 101 + 101 + \cdots + 101$$

Since there are 100 such pairs, $2S = 100 \times 101$, which means $S = \frac{100 \times 101}{2} = 5050$. This principle of pairing terms is a powerful problem-solving strategy.

Practice Problem

An amphitheater has 40 rows of seats. The first row has 18 seats, the second has 20, the third has 22, and so on. Use the principle of symmetry (the Gauss trick) to find the total number of seats in the amphitheater.

(Hint: First, determine the number of seats in the 40th row. Then, pair the first row with the 40th, the second with the 39th, and so on. How many pairs are there, and what is the sum of the seats in each pair?)

Advanced Challenge

Calculate the exact value of the following sum without a calculator:

$$100^2 - 99^2 + 98^2 - 97^2 + \cdots + 2^2 - 1^2$$

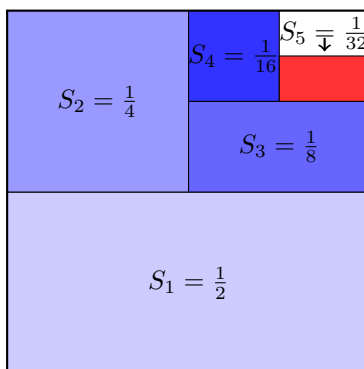
Topic 12: Infinite Series and the Idea of a Limit

Mathematics can sometimes feel like poetry, especially when it touches upon the infinite. For centuries, thinkers were puzzled by paradoxes. Consider one of Zeno's Paradoxes: The swift hero Achilles races a slow tortoise, who is given a 100-meter head start. By the time Achilles runs those 100 meters, the tortoise has moved ahead, say, 10 meters. By the time Achilles covers that 10 meters, the tortoise has moved 1 meter. This continues forever. How can Achilles *ever* overtake the tortoise if he must first cover an infinite number of smaller and smaller distances?

The poetry is in the resolution: an infinite sum of numbers does not have to be infinite. The total distance Achilles must run is $100 + 10 + 1 + 0.1 + \dots$, which is an infinite geometric series. We know this sum converges to a finite number:

$$S = \frac{a}{1-r} = \frac{100}{1-1/10} = \frac{100}{9/10} \approx 111.11 \text{ meters}$$

Achilles does overtake the tortoise. The paradox is resolved by understanding that an infinite process can be completed in a finite amount of time and space. This is the essence of a **limit**.



A square of area 1 is partitioned. The sum of the infinite pieces, $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$, perfectly fills the square, showing the limit of the sum is 1.

Practice Problem

Express the repeating decimal $0.454545\dots$ as a fraction in simplest form.

(Hint: Write the decimal as an infinite series: $0.45 + 0.0045 + 0.000045 + \dots$. This is an infinite geometric series. Find the first term a and the common ratio r , then use the formula for the sum.)

Advanced Challenge

Imagine a point that moves as follows: It travels 1 meter East, then $1/2$ meter North, then $1/4$ meter West, then $1/8$ meter South, then $1/16$ meter East, and so on, continuing the pattern of directions (E, N, W, S) and halving the distance with each step. If this process continues forever, where does the point eventually end up? Find its final (x, y) coordinates, assuming it starts at (0,0).